$$P(x) = \begin{cases} cx^2 & x \in [1,4] \\ 0 & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} P(x) = 1 \Rightarrow \int_{1}^{4} cx^2 = 1$$

$$\Rightarrow c = 1/21$$

Let the CDF be $F(x)$

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} P(t)\,dt \text{ , where } X \text{ is random}$$

- $x < 1$

$$\Rightarrow F(x) = 0 \quad (P(x) = 0$$

- $x \in [1,4]$

$$\Rightarrow F(x) = \int_{1}^{x} \frac{1}{21} t^2 dt = \frac{x^3 - 1}{63}$$

- $x > 4$

$$\Rightarrow F(x) = 1$$